



RAMCO INSTITUTE OF TECHNOLOGY

RAJAPALAYAM - 626 117

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Energy Conservation Quality Circle (ECQC)

&

Department of Electrical and Electronics Engineering

Electrical Energy Conservation Tips in your Home



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Why do we conserve Electrical Energy ?

Reduce Electricity Bill

Conserving Electrical Energy, reduces your Electricity Bill

Reduce CO₂ Emission

Approximately 40% of global CO₂ emissions are emitted from electricity generation through the combustion of fossil fuels to generate heat needed to power steam turbines. Burning these fuels results in the production of carbon dioxide (CO₂)

Save Energy, Save Money, Save the Planet, Save a life



!!! Save Electricity for your better tomorrow !!!

!!! Today's wastage is Tomorrow's shortage !!!

Air Conditioner



Power Rating

1 Ton : **1500 W**

2 Ton : **2000 W**

Time of Operation for 1 Unit Consumption

1 Ton : **40 Minutes**

2 Ton : **24 Minutes**

Energy Conservation Tips

Set the Right default temperature, Keep it at 24 °C, Instead of 18 °C

Do not keep unnecessary things in the AC room

Keep the ceiling fan switched ON at low speed when the AC is running

Make sure your AC room is well-insulated, no crack in wall, no air leak

Regular Servicing and Cleaning

Refrigerator



Power Rating
165 Litre - **225 W**

Time of Operation for
1 Unit Consumption
5 Hours 30 Minutes

Energy Conservation Tips

Replace old refrigerator, buy an energy-efficient refrigerator

Never place your refrigerator next to a heat source such as an oven, radiator, and avoid direct sunlight

Always close the door without delay

Defrost the freezer compartment once in a 15 days

Only place cold food in the refrigerator



Motor Appliances



Power Rating		Time of Operation for 1 Unit Consumption
Fan	- 55 W	18 Hours
Grinder	- 300 W	5 Hours 3 Minutes
Mixer Grinder	- 500 W	2 Hours 6 Minutes
Pump Motor	- 500 W	2 Hours

Energy Conservation Tips

Replace the old motor appliances with energy efficient motors.

Turn it Off, whenever it is not required.

Use Energy Saving Motor Controls

Do regular maintenance to reduce noise and energy consumption.

Size Motors Correctly

Try to use at full load.

Lighting Appliances



Power Rating	Time of Operation for 1 Unit Consumption
Incandescent lamp - 40 W	25 Hours
Fluorescent lamp - 40 W	25 Hours
CFL lamp - 14 W	71 Hours
LED lamp - 10 W	100 Hours

Energy Conservation Tips

Turn-off the lights, that are not being used

Clean the lighting appliances regularly, dust and dirt reduces the illumination

Choose light colour when painting indoor wall, light colour reflects illumination

Replace Incandescent lamp with LED/CFL

Replace ordinary choke with electronic choke in fluorescent lamp

Utilize sun light during day time

Do not keep night lamp ON at all time

Television



Power Rating
120 W

**Time of Operation for
1 Unit Consumption**
8 Hours 20 Minutes

Energy Conservation Tips

Replace the oldest TV with an LED/LCD TV

Turn off the TV by remote control and then unplug the connection, when nobody is watching

Switch Your TV to Energy-Saving Mode

Avoid Watching TV with high brightness and sound

Washing Machine



Power Rating
325 W

**Time of Operation for
1 Unit Consumption**
3 Hours 2 Minutes

Energy Conservation Tips

Replace old washing machine, buy an energy-efficient one.

Wait for a full load, washing machines use the same amount of energy, no matter how big the load.

Avoid using dryers, use sunlight for drying the cloths.

Switch off the power switch at the wall.



Miscellaneous Appliances

Iron Box



Power Rating

1000 Watts

Time of Operation for 1 Unit Consumption

1 Hour

Energy Conservation Tips

Do not spread more water while ironing.

Do not iron the fabrics every day, do it for all fabrics in a week

Water Heater



Power Rating

2000 Watts

Time of Operation for 1 Unit Consumption

30 Minutes

Energy Conservation Tips

Replace your water heater, if it is more than 10 years old

Add thermal insulation to the first 6 feet of both the hot and cold water pipes connected to the heater

Home Appliances Unit Consumption Chart

NAME OF THE APPLIANCES	POWER IN WATTS	TIME PER UNIT
Air Conditioner 1 Ton	1500 W	40 Minutes
Air Conditioner 1.5 Ton	2000 W	24 Minutes
Water Heater	2000 W	30 Minutes
Refrigerator 165 Litre	225 W	5 Hours 30 Minutes
Vacuum Cleaner	750 W	1 Hour 18 Minutes
Pump Water	500 W	2 Hours
Washing Machine	325 W	3 Hours 2 Minutes
Mixer Grinder	500 W	2 Hours
Grinder	300 W	3 Hours 20 Minutes
Fan	40 W	25 Hours
Air Cooler	170 W	5 Hours 50 Minutes
Iron Box	1000 W	1 Hour
Fluorescent Lamp	40 W	25 Hours
Incandescent Lamp	40 W	25 Hours
Night Lamp	15 W	66 Hours
Television	120 W	8 Hours 20 Minutes
Tape Recorder	20 W	50 Hours
Electric Cooker	1000 W	1 Hour 18 Minutes

Ways to conserve Electrical Energy?

Replace your incandescent Lamp with CFL or LED bulb

Replace inefficient appliances with Energy Star Appliances

Turn-off the computer, laptop after use, especially overnight

Turn-off the appliances, when not in use

Clean the indoor wall and lighting appliances regularly

Unplug devices, when not in use

Keep thermostat at low temperature

Use sun light at day time for illumination and drying cloths

Do not leave water heater ON for a long time



!!! Turn off the light, Keep the future bright. !!!

!!! Conservation: It doesn't cost. It saves. !!!

Electricity Tariff Rates in Tamilnadu

Revised Tariff rates with effect from 11.08.2017

It is applicable to the low tension supply consumer

Consumption upto 100 units bi-monthly

Unit Consumption	Fixed Charge	Energy Charges
1 – 100 Units	Rs. 0	Rs. 0

Consumption above 100 units and upto 200 units bi-monthly

Unit Consumption	Fixed Charge	Energy Charges
1 – 100 Units	Rs. 20	Rs. 0
101 – 200 Units		Rs. 1.5 per Unit

Consumption above 200 units and upto 500 units bi-monthly

Unit Consumption	Fixed Charge	Energy Charges
1 – 100 Units	Rs. 30	Rs. 0
101 – 200 Units		Rs. 2 per Unit
201 – 500 Units		Rs. 3 per Unit

Consumption above 500 units

Unit Consumption	Fixed Charge	Energy Charges
1 – 100 Units	Rs. 50	Rs. 0
101 – 200 Units		Rs. 3.50 per Unit
201 – 500 Units		Rs. 4.60 per Unit
Above 500 Units		Rs. 6.60 per Unit